



2019 Year 12 Course Assessment Task 4 (Trial Examination)
Monday August 12, 2019

- Working time – 3 hours.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided (on page 11)

SECTION II

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: **# BOOKLETS USED:**

Class (please ✓)

○ 12MAT.3 – Mrs Gan

○ 12MAT.4 – Ms Park

○ 12MAT.5 – Mr Sekaran

○ 12MAT.6 – Mrs Dupuche

Marker's use only.

[illegible]

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 10 minutes for this section

Mark your answers on the answer grid provided (labelled as page 11).

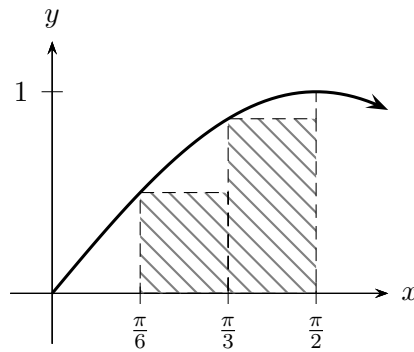
Questions

Marks

1. An experiment consists of tossing a coin and then rolling a fair six-sided die. What is the probability of observing a 'head' and a 'six'? 1
- (A) $\frac{1}{2}$ (B) $\frac{1}{35}$ (C) $\frac{7}{12}$ (D) $\frac{1}{12}$
2. What is the solution to the equation $\log_3(2x - 5) = 2$? 1
- (A) $x = 1$ (B) $x = \frac{13}{2}$ (C) $x = \frac{11}{2}$ (D) $x = 7$
3. What is the derivative of $\frac{\sin x}{\cos x + 1}$? 1
- (A) $\frac{\sin x + \cos x}{\sin^2 x}$ (C) $\frac{1}{\cos x + 1}$
- (B) $\frac{\cos^2 x + \cos x - \sin^2 x}{(\cos x + 1)^2}$ (D) $\frac{2}{\cos x + 1}$
4. What is the result of evaluating $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x - 2}$? 1
- (A) 5 (B) 6 (C) 7 (D) 8
5. What are the coordinates of the turning point to the curve $y = e^x - ex$? 1
- (A) (0, 1) (B) (1, 0) (C) (1, e) (D) (e, 1)

6. The area beneath the curve $y = \sin x$ between $x = 0$ and $x = \frac{\pi}{2}$ is approximated by the two rectangles as shown.

1



What is the approximation to the area?

- (A) $\frac{\pi}{2}$ square units
 (B) $\frac{2\pi}{3}$ square units
 (C) $\frac{(1 + \sqrt{3})\pi}{12}$ square units
 (D) $\frac{(1 + \sqrt{3})\pi}{6}$ square units

7. The first three terms of an arithmetic series are 5, 9 and 13.

1

What is the 15th term of the series?

- (A) 61 (B) 66 (C) 495 (D) 585

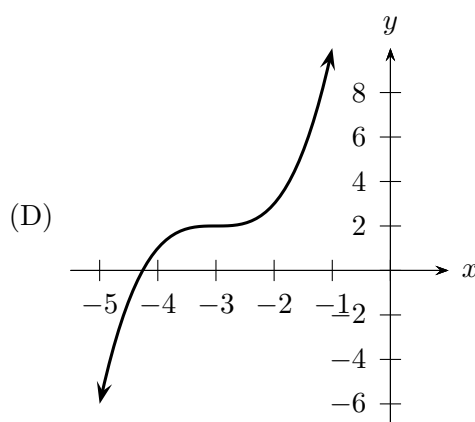
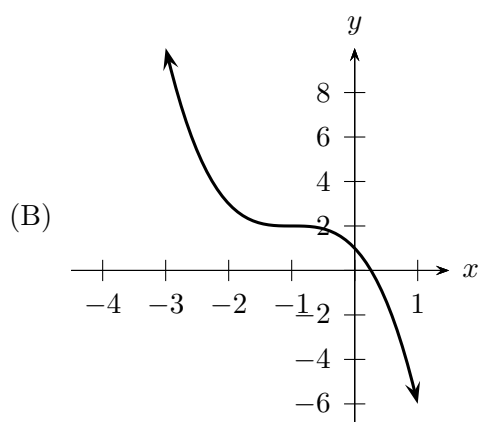
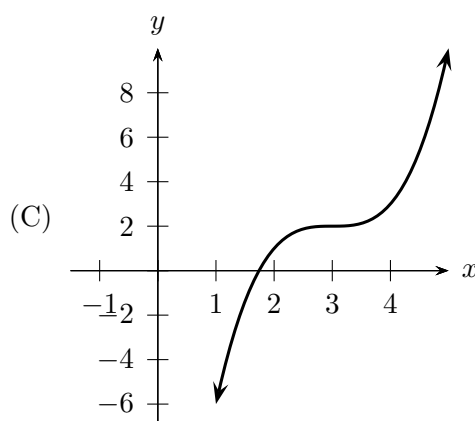
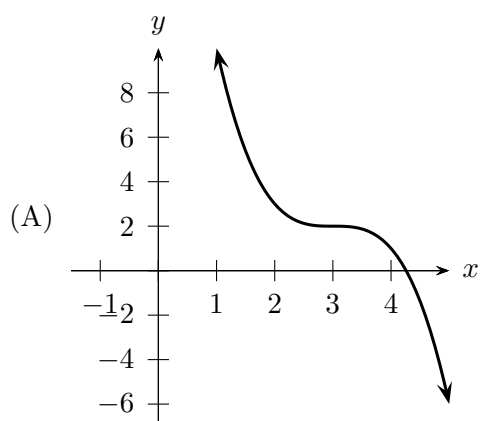
8. What is the solution to $|x - 2| < 3$?

1

- (A) $-1 < x < 5$
 (B) $x < -5$ or $x > 1$
 (C) $-5 < x < 1$
 (D) $x < -1$ or $x > 5$

9. Which diagram is the graph of $y = 2 - (x - 3)^3$?

1



10. For what values of x does the geometric progression $1 + \frac{1}{2-x} + \frac{1}{(2-x)^2} + \dots$ has a limiting sum?

1

(A) $1 < x < 3$

(C) $x \neq 2$

(B) $x < 1$ or $x > 3$

(D) $0 < x < 2$

Examination continues overleaf...

Section II

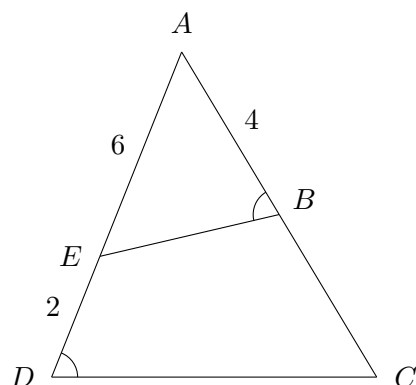
90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours and 50 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

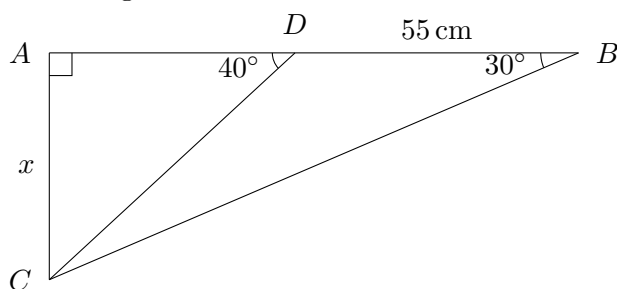
Question 11 (15 Marks)	Commence a NEW booklet.	Marks
(a) Find integers a and b such that $\frac{4 + \sqrt{3}}{2 + \sqrt{3}} \equiv a\sqrt{3} + b$.		2
(b) Find the domain of the function $y = \frac{1}{x^2 - 4}$.		2
(c) Find $\int \frac{1}{(x + 5)^3} dx$.		2
(d) $\triangle ABE$ is similar to $\triangle ACD$. $AE = 6$, $AB = 4$ and $ED = 2$.		2



Find the length of BC .

- (e) There are 30 students in a class with 24 students exercising by running and 18 exercising at the gym. Each student does at least one type of exercise.
- One student is chosen at random. What is the probability that the student exercises at the gym? **1**
 - Two students are chosen at random. What is the probability that both students exercise at the gym? **1**
 - Two students are chosen at random. What is the probability that both students do both types of exercise? **1**
 - One student is chosen at random. What is the probability that the student does *not* do both types of exercise? **1**

- (f) Find the value of x in the diagram below, where $BD = 55$ cm. Give your answer correct to 3 significant figures. **3**

**Question 12** (15 Marks)

Commence a NEW booklet.

Marks

- (a) The line ℓ_1 passes through the point $(9, -4)$ and has gradient of $\frac{1}{3}$.
- Find the equation of ℓ_1 in the form $ax + by + c = 0$, where a , b and c are integers. **2**
 - Line ℓ_2 passes through the origin and has gradient -2 . **2**
- The lines ℓ_1 and ℓ_2 intersect at the point P . What are the coordinates of P ?
- Given that ℓ_1 crosses the y axis at the point C . **2**
- Calculate the exact area of $\triangle OPC$.
- (b) Differentiate with respect to x :
- $(4x^3 - x)^7$. **1**
 - $e^x \cos 4x$. **2**
 - $\frac{x^2 + 2}{3x - 4}$ **2**
- (c) Consider the parabola $4y = x^2 - 2x + 5$.
- Find the coordinates of the vertex. **2**
 - Find the coordinates of the focus. **1**
 - Find the values of x such that the function $4y = x^2 - 2x + 5$ is increasing. **1**

Question 13 (15 Marks)		Commence a NEW booklet.	Marks
(a)	i. What is the period of the function $y = 4 \sin 2x$?		1
	ii. Sketch the function $y = 4 \sin 2x + 1$ for $-\pi \leq x \leq \pi$.		2
(b)	i. Show that $\frac{x+2}{5x^2+7x-6} = \frac{1}{5x-3}$		1
	ii. Hence find the value of k such that		3
		$\int_1^k \frac{x+2}{5x^2+7x-6} dx = \frac{1}{5} \ln 6$	
(c)	A curve has gradient function with equation		
		$\frac{dy}{dx} = 6(x-1)(x-2)$	
	i. If the curve passes through the point $(0, -3)$, what is the equation of the curve?		2
	ii. Find the coordinates of the stationary points.		1
	iii. Determine the nature of the stationary points.		1
	iv. Find the coordinates of the point of inflexion.		2
	v. Sketch the curve, showing the essential features.		2

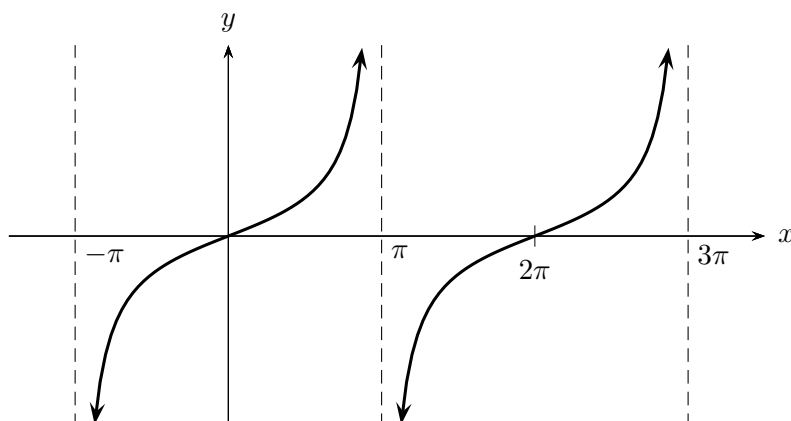
Examination continues overleaf...

Question 14 (15 Marks)

Commence a NEW booklet.

Marks

- (a) The graph of $f(x) = \tan \frac{x}{2}$ from $x = -\pi$ to $x = 3\pi$ is shown below.



- i. Find $f' \left(\frac{\pi}{2} \right)$. **1**
 - ii. Find the equation of the normal to the graph of $y = f(x)$ at the point where $x = \frac{\pi}{2}$. **2**
 - iii. Find all the points on the graph of $y = f(x)$ in the given domain, where the gradient equals to 1. **2**
- (b) i. Copy and complete the table below for $y = \sqrt{5^x + 2}$ in your writing booklet. **1**

x	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2
y			2.646		

- ii. Use the Trapezoidal Rule to find an approximation for the value of **2**

$$\int_0^2 \sqrt{5^x + 2} \, dx$$

- (c) The second and fifth terms of a geometric series are 750 and -6 respectively.
- i. Find the common ratio, and the first term of the series. **2**
 - ii. What is the limiting sum of the series? **1**
- (d) The quadratic equation $3x^2 + 9x + 1 = 0$ has roots α and β . Find the values of:
- i. $\alpha + \beta$. **1**
 - ii. $\alpha\beta$. **1**
 - iii. $4\alpha\beta^2 + 4\alpha^2\beta$. **1**
 - iv. $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$. **1**

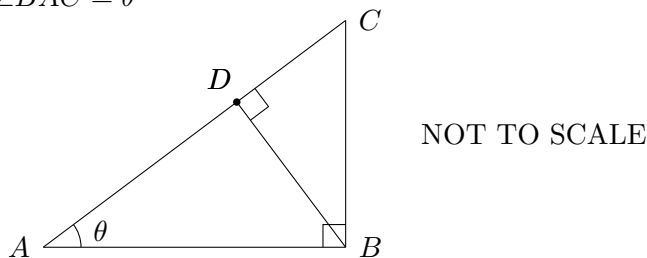
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Question 15 (15 Marks)

Commence a NEW booklet.

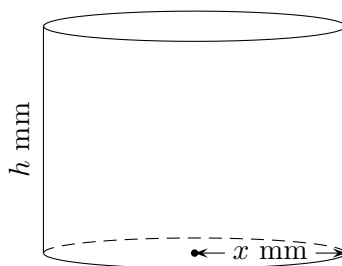
Marks

- (a) $\triangle ABC$ is right angled at B . D is a point on AC such that BD is perpendicular to AC . Let $\angle BAC = \theta$



You are given that $6AD + BC = 5AC$.

- i. Show that $6 \cos \theta + \tan \theta = 5 \sec \theta$ **1**
 - ii. Deduce that $6 \sin^2 \theta - \sin \theta - 1 = 0$ **1**
 - iii. Find the value of θ , correct to the nearest degree. **2**
- (b) A closed cylinder with base radius x millimetres and height h millimetres is shown. The volume of the cylinder is 60 mm^3 .



- i. Find an expression for h in terms of x . **1**
 - ii. Show that the surface area $A \text{ mm}^2$ of the cylinder is given by **1**
- $$A = \frac{120}{x} + 2\pi x^2$$
- iii. Find the value of x which minimises the surface area of the cylinder. **3**
 - iv. Calculate the minimum surface area of the cylinder, correct to the nearest square millimetres. **1**

- (c) A biased coin is tossed three times. The probability of obtaining a head is

$$P(H) = \frac{2}{5}$$

- i. Find the probability of obtaining two heads and a tail. **1**
 - ii. Find the probability of obtaining at least one head. **1**
- (d) Find the value of m such that the curves $y = \frac{x^4}{m}$ and $y = -3x^3 - 2x^2$ intersect at more than two points. **3**

Examination continues overleaf...

Question 16 (15 Marks)

Commence a NEW booklet.

Marks

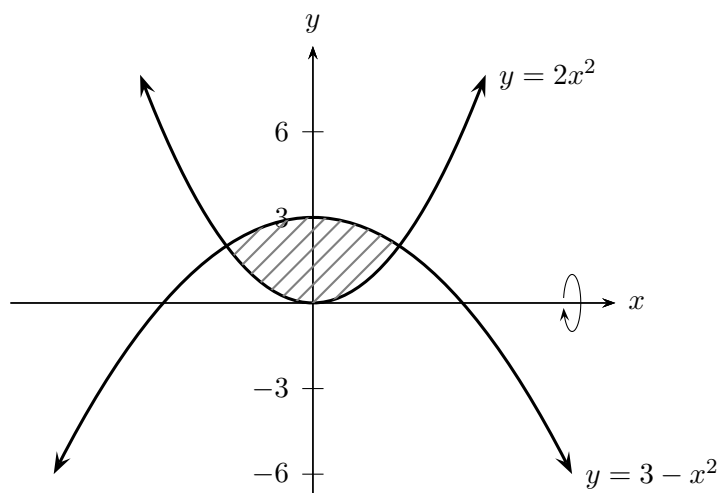
- (a) Aiden borrows \$15 000 to purchase solar panels for his home. The interest is calculated monthly at the rate of 6% per annum, and is compounded monthly. He intends to repay the loan in monthly instalments of \$ M .
- How much does Aiden owe at the end of the first month, before he makes his first repayment? **1**
 - Let A_n be the amount of money owing after n repayments. Show that when $n = 3$, **2**

$$A_3 = (15\,000 \times 1.005^3) - M(1 + 1.005 + 1.005^2)$$

- After two years of repaying the loan, Aiden still owes \$10 000 on the loan. **3**

What is the amount of the monthly repayment?

- (b) The graphs of $y = 2x^2$ and $y = 3 - x^2$ are shown.



- Find the coordinates of the points of intersection between the two curves. **1**
 - The shaded region is rotated about the x axis. Find the *exact* volume of the solid formed. **3**
- (c) On a factory production line, a tap opens and closes to fill containers with liquid. As the tap opens, the rate of flow R increases for the first 10 seconds according to the rule $R = \frac{3t}{25}$, where R is measured in litres per second.

The rate of flow then remains constant until the tap begins to close.

- Show that while the tap is still fully open, the volume in the container at any time is given by **3**

$$V = \frac{6}{5}(t - 5)$$

- For how many seconds must the tap remain fully open in order to exactly fill a 120 L container without any spillage? **2**

End of paper.

2019 HSC Trial Mathematics (2U) Solutions

Section 1

1. 2. 3. 4. 5. 6. 7. 8. 9. 10.
D D C A B C A A A B

Question 11

$$a) \frac{4+\sqrt{3}}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}}$$

$$= \frac{(4+\sqrt{3})(2-\sqrt{3})}{4-3}$$

$$= 8 - 2\sqrt{3} - 3$$

$$= 5 - 2\sqrt{3}$$

$$\therefore a = -2$$

$$b = 5$$

$$b) x^2 - 4 \neq 0$$

$$x \neq \pm 2$$

$$D: \{x \in \mathbb{R}, x \neq \pm 2\}$$

$$c) \int (x+5)^{-3} dx$$

$$= \frac{(x+5)^{-2}}{-2} + C$$

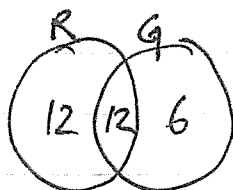
$$= -\frac{1}{2(x+5)^2} + C$$

$$d) \frac{BC+4}{6} = \frac{2+6}{4}$$

$$BC+4 = 12$$

$$BC = 8$$

e)



$$i) \frac{18}{30} = \frac{3}{5}$$

$$ii) \frac{18}{30} \times \frac{17}{29} = \frac{51}{145}$$

Q11 (cont)

$$e) iii) \frac{12}{30} \times \frac{4}{29} = \frac{22}{145} \quad \checkmark \quad (1)$$

$$iv) 1 - \frac{12}{30} = \frac{3}{5} \quad \checkmark \quad (1)$$

$$f) \angle DCB = 40^\circ - 30^\circ = 10^\circ$$

In $\triangle BCD$,

$$\frac{DC}{\sin 30^\circ} = \frac{55}{\sin 10^\circ} \quad \checkmark$$

$$DC = \frac{55}{2 \sin 10^\circ}$$

In $\triangle ADC$,

$$\sin 40^\circ = \frac{x}{DC} \quad \checkmark$$

$$x = \frac{55 \sin 40^\circ}{2 \sin 10^\circ}$$

$$= 101.7958...$$

$$= 102 \text{ cm (3 sig fig.)} \quad \checkmark \quad (3)$$

Question 12

$$a) i) y - (-4) = \frac{1}{3}(x - 9) \quad \checkmark$$

$$3y + 12 = x - 9$$

$$L: x - 3y - 21 = 0 \quad \checkmark \quad (2)$$

$$ii) L_2: y = -2x$$

$$x - 3(-2x) - 21 = 0 \quad \checkmark$$

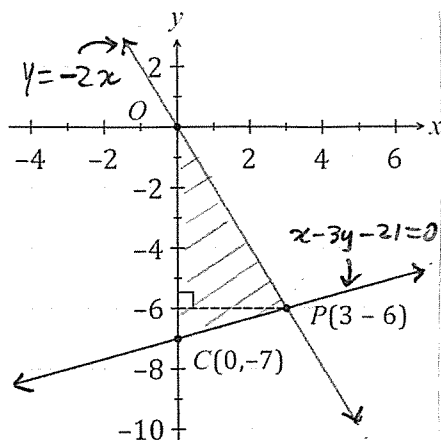
$$7x = 21$$

$$x = 3$$

$$\therefore P \text{ is } (3, -6) \quad \checkmark \quad (2)$$

Q12 (cont)

a) iii)



$$A+C, \quad -3y-21=0$$

$$y = -7$$

$$\therefore C(0, -7)$$

Area of $\triangle OPC$

$$= \frac{1}{2} \times 7 \times 3$$

$$= 10\frac{1}{2} \text{ sq. units}$$

b) i) $\frac{d}{dx}(4x^3 - x)^7$

$$= 7(4x^3 - x)^6(12x^2 - 1)$$

ii) $\frac{d}{dx}(e^x \cos 4x)$

$$= e^x(-4\sin 4x) + (\cos 4x)e^x$$

$$= e^x(\cos 4x - 4\sin 4x)$$

iii) $\frac{d}{dx}\left(\frac{x^2+2}{3x-4}\right)$

$$= \frac{(3x-4)(2x) - (x^2+2)(3)}{(3x-4)^2}$$

$$= \frac{6x^2 - 8x - 3x^2 - 6}{(3x-4)^2}$$

$$= \frac{3x^2 - 8x - 6}{(3x-4)^2}$$

c) i) $4y = x^2 - 2x + 5$

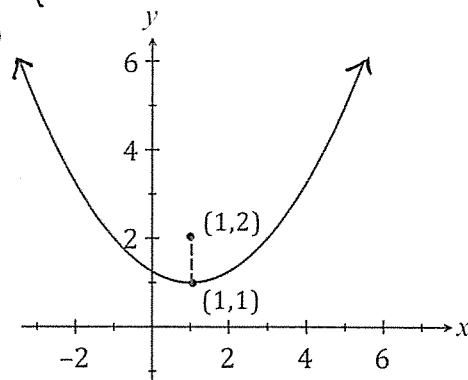
$$x^2 - 2x = 4y - 5$$

$$(x-1)^2 = 4(y-1)$$

$\therefore$ vertex is $(1, 1)$

ii) focus is $(1, 2)$

iii)

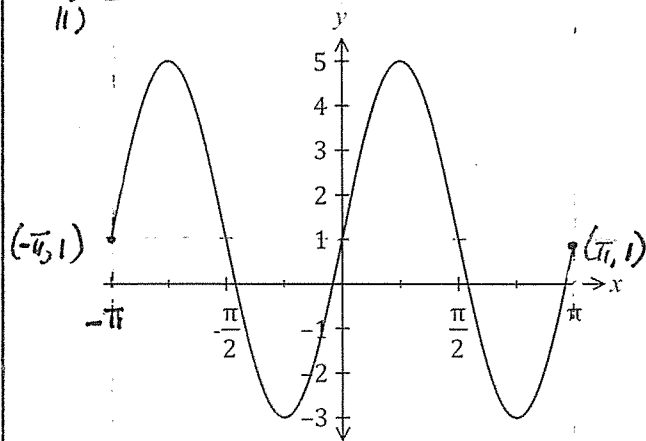


$$x > 1$$

Question 13

a) i) π

ii)



shape

y intercept
end points } labelled

Q13 (cont)

$$\text{b) i) } \frac{x+2}{5x^2+7x-6} = \frac{x+2}{(5x-3)(x+2)} \quad \checkmark$$

$$= \frac{1}{5x-3} \quad \textcircled{1}$$

$$\text{ii) } \int_1^k \frac{x+2}{5x^2+7x-6} dx$$

$$= \frac{1}{5} \int_1^k \frac{5}{5x-3} dx$$

$$= \frac{1}{5} [\ln(5x-3)]_1^k \quad \checkmark$$

$$= \frac{1}{5} [\ln(5k-3) - \ln 2]$$

$$= \frac{1}{5} \ln \frac{5k-3}{2}$$

$$\frac{5k-3}{2} = 6 \quad \checkmark$$

$$5k-3=12$$

$$k=3 \quad \checkmark \quad \textcircled{3}$$

$$\text{c) i) } \frac{dy}{dx} = 6(x-1)(x-2)$$

$$= 6x^2 - 18x + 12$$

$$y = \int (6x^2 - 18x + 12) dx$$

$$= 2x^3 - 9x^2 + 12x + C \quad \checkmark$$

$$\text{Sub } (0, -3),$$

$$-3 = C$$

$$\therefore y = 2x^3 - 9x^2 + 12x - 3 \quad \checkmark \quad \textcircled{2}$$

$$\text{ii) } (1, 2), (2, 1) \quad \checkmark \quad \textcircled{1}$$

$$\text{iii) } \frac{d^2y}{dx^2} = 12x - 18$$

$$\text{At } (1, 2), \frac{d^2y}{dx^2} < 0$$

$$\text{At } (2, 1), \frac{d^2y}{dx^2} > 0 \quad \textcircled{1}$$

$\therefore (1, 2)$ is a maximum turning point $\checkmark$
and $(2, 1)$ is a minimum turning point.

$$\text{iv) } 12x - 18 = 0$$

$$x = 1\frac{1}{2}$$

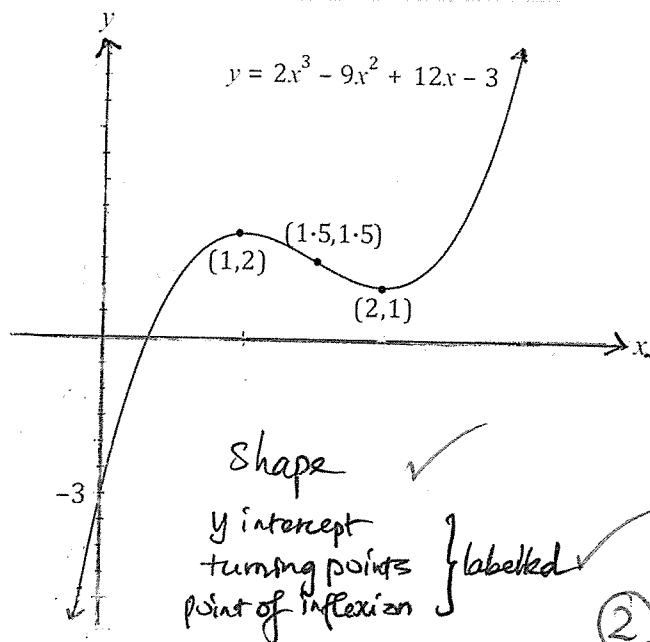
$$y = 1\frac{1}{2}$$

x	1.4	1.5	1.6
$\frac{d^2y}{dx^2}$	-1.2	0	1.2

$\frac{d^2y}{dx^2}$ changes sign around $x = 1.5$ $\checkmark$

$\therefore (1\frac{1}{2}, 1\frac{1}{2})$ is a point of inflexion. $\checkmark \quad \textcircled{2}$

v)



Question 14

a) i) $f(x) = \tan \frac{x}{2}$

$$f'(x) = \frac{1}{2} \sec^2\left(\frac{x}{2}\right)$$

$$f'\left(\frac{\pi}{2}\right) = \frac{1}{2} \sec^2\left(\frac{\pi}{4}\right)$$

$$= 1$$

ii) $x = \frac{\pi}{2}, y = 1$

$\therefore$ equation of normal is:

$$y - 1 = -1\left(x - \frac{\pi}{2}\right)$$

$$y = -x + \frac{\pi}{2} + 1$$

iii) $f'(x) = 1$

$$\frac{1}{2} \sec^2\left(\frac{x}{2}\right) = 1$$

$$\sec^2\left(\frac{x}{2}\right) = 2$$

$$\cos \frac{x}{2} = \pm \frac{1}{\sqrt{2}}$$

in the points are:

$$\left(-\frac{\pi}{2}, -1\right), \left(\frac{\pi}{2}, 1\right), \left(\frac{3\pi}{2}, -1\right), \left(\frac{5\pi}{2}, 1\right)$$

b) ii)

x	0	0.5	1	1.5	2
y	1.732	2.058	2.646	3.630	5.196

$$\text{ii) } 0.5 \left[1.732 + 5.196 + 2(2.058 + 2.646 + 3.630) \right]$$

$$= 5.899$$

c) ar = 750 — ①

ar = -6 — ②

$$\frac{②}{①}, t^3 = -\frac{1}{125}$$

$$t = -\frac{1}{5}$$

Sub into ①,

$$a = -3750$$

$$\text{ii) } \frac{-3750}{1 - (-\frac{1}{5})}$$

$$= -3125$$

d) i) $-\frac{9}{3} = -3$

ii) $\frac{1}{3}$

iii) $4\alpha\beta^2 + 4\alpha^2\beta$

$$= 4\alpha\beta(\beta + \alpha)$$

$$= 4 \times \frac{1}{3} \times (-3)$$

$$= -4$$

iv) $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$$

$$= \frac{(-3)^2 - 2(\frac{1}{3})}{(\frac{1}{3})^2}$$

$$= 75$$

Question 15

a) In $\triangle ABD$, $\cos \theta = \frac{AD}{AB}$

i) $\therefore AD = AB \cos \theta$

In $\triangle ABC$, $\tan \theta = \frac{BC}{AB}$

$$\therefore BC = AB \tan \theta$$

$$\sec \theta = \frac{AC}{AB}$$

$$\therefore AC = AB \sec \theta$$

$$6AD + BC = 5AC$$

$$\therefore 6AB \cos \theta + AB \tan \theta = 5AB \sec \theta$$

$$6 \cos \theta + \tan \theta = 5 \sec \theta$$

ii) $6 \cos \theta + \frac{\sin \theta}{\cos \theta} = \frac{5}{\cos \theta}$

$$6 \cos^2 \theta + \sin \theta = 5$$

$$6(1 - \sin^2 \theta) + \sin \theta = 5$$

$$6 \sin^2 \theta - \sin \theta - 1 = 0$$

iii) $\sin \theta = \frac{1 \pm \sqrt{1 + 4(6)}}{2(6)}$

$$= \frac{1 \pm 5}{12}$$

$$= \frac{1}{2} \text{ or } -\frac{1}{3}$$

$$\theta = 30^\circ \text{ as } 0 \leq \theta \leq 90^\circ$$

Q15 (cont)

b) i) $60 = \pi x^2 h$

$$h = \frac{60}{\pi x^2}$$

ii) $A = 2\pi r h + 2\pi r^2$

$$= 2\pi x \times \frac{60}{\pi x^2} + 2\pi x^2$$

$$= \frac{120}{x} + 2\pi x^2$$

iii) $\frac{dA}{dx} = -120x^{-2} + 4\pi x$
 $= 0$

when $4\pi x = \frac{120}{x^2}$

$$x = \sqrt[3]{\frac{30}{\pi}}$$

$$\frac{d^2A}{dx^2} = 240x^{-3} + 4\pi$$

$$= 240 \times \frac{\pi}{30} + 4\pi$$

$$= 12\pi > 0$$

$\therefore A$ is minimum when

$$x = \sqrt[3]{\frac{30}{\pi}}$$

iv) $\text{Min } A = 120 \times \sqrt[3]{\frac{30}{\pi}} + 2\pi \left(\sqrt[3]{\frac{30}{\pi}}\right)^2$

$$= 84.8428 \dots$$

$$\div 85 \text{ mm}^2$$

c) i) $P(HTT) + P(HTH) + P(THH)$

$$= \frac{2}{5} \times \frac{2}{5} \times \frac{3}{5} \times 3$$

$$= \frac{36}{125}$$

ii) $P(1H)$

$$= 1 - (TTT)$$

$$= 1 - \frac{3}{5} \times \frac{3}{5} \times \frac{3}{5}$$

$$= \frac{98}{125}$$

d) $\frac{x^4}{m} = -3x^3 - 2x^2$

$$x^4 + 3mx^3 + 2mx^2 = 0$$

$$x^2(x^2 + 3mx + 2m) = 0$$

$x = 0$ is a solution

For two more solutions,

$$x^2 + 3mx + 2m = 0$$

$$\Delta \geq 0$$

$$9m^2 - 4(1)(2m) > 0$$

$$m(9m - 8) > 0$$

$$m < 0 \text{ or } m > \frac{8}{9}$$

Question 16

a) i) 15000×1.005
 $= \$15075$

ii) $A_1 = 15000 \times 1.005 - M$

$$A_2 = (15000 \times 1.005 - M) \times 1.005 - M$$

$$= 15000 \times 1.005^2 - M(1 + 1.005)$$

$$A_3 = [15000 \times 1.005^2 - M(1 + 1.005)] \times 1.005 - M$$

$$= 15000 \times 1.005^3 - M(1 + 1.005 + 1.005^2)$$

iii) $A_{24} = 15000 \times 1.005^{24} - M(1 + 1.005 + \dots + 1.005^{23})$

$$10000 = 15000 \times 1.005^{24} - M(1 + 1.005 + \dots + 1.005^{23})$$

$$= 15000 \times 1.005^{24} - M \times \frac{1(1.005^{24} - 1)}{1.005 - 1}$$

$$M = \frac{(15000 \times 1.005^{24} - 10000) \times 0.005}{1.005^{24} - 1}$$

$$= \$271.60$$

Q16 cont

$$b) i) 2x^2 = 3 - x^2$$

$$3x^2 = 3$$

$$x = \pm 1$$

$\therefore$ points are intersection are

$$(-1, 2), (1, 2)$$

$$ii) V = 2\pi \int_0^1 [(3-x^2)^2 - (2x^2)^2] dx$$

$$= 2\pi \int_0^1 (9 - 6x^2 + x^4 - 4x^4) dx$$

$$= 2\pi \int_0^1 (-3x^4 - 6x^2 + 9) dx$$

$$= 2\pi \left[-\frac{3x^5}{5} - 2x^3 + 9x \right]_0^1$$

$$= 2\pi \left(-\frac{3}{5} - 2 + 9 \right)$$

$$= \frac{64\pi}{5} \text{ m. units}$$

Question 16 c i

$$\text{When } 0 < t \leq 10, \quad \frac{dV}{dt} = \frac{3}{25}t$$

$$\text{So,} \quad V = \frac{1}{2} \times \frac{3}{25}t^2 + c_1$$

$$\text{And} \quad t = 0, V = 0 \Rightarrow c_1 = 0$$

$$\therefore V = \frac{3}{50}t^2$$

$$\text{When } t = 10, \quad V = \frac{3}{50}(10)^2 = 6 \text{ L}$$

$$\text{And when } t \geq 10, \quad \frac{dV}{dt} = \frac{3}{25} \times 10 = \frac{6}{5} \text{ L/s}$$

(actually $10 \leq t < t_1$
if you consider part ii)

$$\text{So when } t \geq 10, \quad V = \frac{6}{5}t + c_2$$

(actually $10 \leq t < t_1$
if you consider part ii)

$$\text{And} \quad t = 10, V = 6 \Rightarrow c_2 = -6$$

$$\therefore V = \frac{6}{5}t - 6 = \frac{6}{5}(t - 5)$$

as required.

Alternatively (1):

$$V = \int_0^{10} \frac{3t}{25} dt$$

$$= \frac{3}{25} \int_0^{10} t dt$$

$$= 6 \text{ L}$$

$$\text{When } t > 10, \quad \frac{dV}{dt} = \frac{3}{25} \times 10 = \frac{6}{5} \text{ L/s}$$

(actually $10 \leq t < t_1$
if you consider part ii)

$$\text{So when } t > 10, \quad V = V_{\text{up to } 10\text{s}} + V_{\text{after } 10\text{s}}$$

(actually $10 \leq t < t_1$
if you consider part ii)

$$= 6 + \frac{dV}{dt} \times t$$

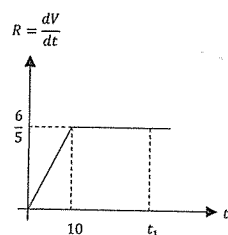
$$= 6 + \frac{6}{5} \times (t - 10)$$

$$= 6 + \frac{6}{5}t - 12$$

$$= \frac{6}{5}t - 6$$

$$= \frac{6}{5}(t - 5)$$

Alternatively (2):



$$V = \int_0^t R dt$$

$$= \text{Area under } R(t)$$

$$= \frac{6}{5}t - \frac{1}{2} \left(\frac{6}{5} \times 10 \right)$$

$$= \frac{6}{5}t - 6$$

$$= \frac{6}{5}(t - 5)$$

Alternatively (3):

$$V = \frac{6}{5}t - \int_0^{10} R dt$$

$$= \frac{6}{5}t - \int_0^{10} \frac{3}{25}t dt$$

$$= \frac{6}{5}t - 6$$

$$= \frac{6}{5}(t - 5)$$

Question 16 c ii

Two ways you can fill the container:

1. Place container under closed tap $\rightarrow$ open tap $\rightarrow$ close the tap to fill to 120 L

t	0 s	10 s	t ₁ s	t ₂ s
Tap is	Closed	Opening	Fully open	Closing
$\frac{dV}{dt} = R$	0	$\frac{3t}{25}$	$\frac{6}{5}$	$-\frac{3t}{25}$
V	0 L	6 L	120 - 6 = 114 L	120 L

$$\text{Solving for when } V = 114 \text{ L,} \quad \left. \begin{aligned} 114 &= \frac{6}{5}(t - 5) \\ 95 &= t - 5 \\ t &= 100 \end{aligned} \right\} \checkmark$$

$\therefore$ Tap has been fully open for $100 - 10 = 90 \text{ s}$ $\checkmark$

2. Place container under closed tap $\rightarrow$ open tap $\rightarrow$ remove container from under the tap when it reaches 120 L

t	0 s	10 s	t ₁ s	
Tap is	Closed	Opening	Fully open	
$\frac{dV}{dt} = R$	0	$\frac{3t}{25}$	$\frac{6}{5}$	
V	0 L	6 L	120 L	

$$\text{Solving for when } V = 120 \text{ L,} \quad \left. \begin{aligned} 120 &= \frac{6}{5}(t - 5) \\ 100 &= t - 5 \\ t &= 105 \end{aligned} \right\} \checkmark$$

$\therefore$ Tap has been fully open for $105 - 10 = 95 \text{ s}$ $\checkmark$